📊 EMI Prediction System using Machine Learning

1. Introduction

The EMI Prediction System is a machine learning-based application designed to assess loan eligibility and predict the maximum monthly EMI a user can afford. This system helps financial institutions make data-driven decisions while also assisting users in understanding their borrowing capacity.

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2. Problem Statement

Loan approval processes often involve manual evaluation and are prone to bias and inefficiency. Additionally, determining an applicant's EMI affordability is complex due to multiple influencing factors.

This project aims to:

- Classify applicants into loan eligibility categories
- Predict the maximum EMI a user can handle

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3. Objectives

- Build a **classification model** to categorize applicants:
  - Eligible
  - High Risk
  - Not Eligible
- Build a **regression model** to predict:
  - Maximum Monthly EMI
- Handle **imbalanced dataset issues**
- Compare multiple ML models
- Track experiments using MLflow
- Deploy using Streamlit

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4. Dataset Description

The dataset contains applicant financial and personal details such as:

- Income
- Age
- Employment Type
- Existing Loans

- Credit Score
- Dependents
- Loan Amount

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5. Data Preprocessing

Steps performed:

- Handling missing values
- Encoding categorical variables
- Feature engineering
- Scaling numerical features
- Splitting into train, validation, and test sets

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6. Feature Engineering

Custom features were created to improve model performance:

- Income-to-loan ratio
- EMI affordability indicators
- Risk-based derived features

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7. Handling Imbalanced Data

The dataset was imbalanced, so:

- SMOTE (Synthetic Minority Oversampling Technique) was used
- Controlled oversampling was applied to avoid overfitting

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8. Model Building

8.1 Classification Models

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier

Final Model: XGBoost (Best Recall for Risk Class ~0.86)

8.2 Regression Models

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor

Final Model: XGBoost Regressor

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9. Model Evaluation

Classification Metrics:

- Accuracy
- Precision
- Recall (Focus on minority class)
- Confusion Matrix

Regression Metrics:

- RMSE
- R^2 Score

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10. Experiment Tracking

MLflow was used to:

- Track model performance
- Compare multiple runs
- Log parameters and metrics

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11. Deployment

Streamlit Application Features:

- User input form
- Loan eligibility prediction
- EMI prediction
- Real-time results display

Models Used:

- Classification Model: xgb_classifier_final.pkl
- Regression Model: xgb_regressor_final.pkl

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12. Challenges Faced

- Handling imbalanced dataset
- Feature selection and engineering
- Model overfitting
- Integration of classification and regression pipelines
- Bug fix: Separate feature columns for classification and regression models

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13. Results

- Classification model achieved strong recall for high-risk class (~0.86)
- Regression model showed good accuracy in EMI prediction
- End-to-end pipeline successfully deployed

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14. Conclusion

The EMI Prediction System successfully demonstrates how machine learning can automate loan decision-making and provide financial insights. The system is scalable and can be enhanced with more real-world financial data.

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15. Future Improvements

- Add real-time credit score integration
- Improve dataset size and quality
- Deploy on cloud platform
- Add explainability (SHAP/LIME)
- Enhance UI/UX

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16. Tech Stack

- Python
- Pandas, NumPy
- Scikit-learn
- XGBoost
- MLflow
- Streamlit

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17. How to Run the Project

1. Install dependencies
2. Load trained models
3. Run Streamlit app:
4. streamlit run app.py
5. Enter user details and view predictions

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18. Author

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Machine Learning Project – EMI Prediction System